

Radar-based Sentiment Analysis and Recognition

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Abstract

1 Introduction

Sentiment analysis commonly refers to the process of identifying and extracting subjective information from text, speech, or other forms of data [Ain *et al.*, 2017]. In the context of this paper, we focus on sentiment analysis using radar-based data. Radar sensors are a powerful tool to capture subtle movements and gestures while being non-intrusive, privacy-preserving and much easier to deploy in real-world scenarios compared to other modalities such as cameras or wearable sensors [Sun *et al.*, 2020]. The data from these sensors have proved to be useful in applications such as gesture recognition [Sun *et al.*, 2020], object detection [Manjunath *et al.*, 2018] and human activity recognition [Froehlich *et al.*, 2024].

However, the use of radar-based data for sentiment analysis and recognition is still a relatively unexplored area, with limited research and few datasets available. The characteristics of radar signals, such as their ability to capture micro-movements and gestures, make them a promising source of information for sentiment analysis [Sun *et al.*, 2020].

The objective of this paper is to explore the potential of radar-based data for sentiment analysis and recognition. We propose different approaches to extract meaningful features from radar signals and evaluate how effective the classification models are in recognizing different sentiments.

2 Related Work

This research is based on the experiments conducted by [Zuo *et al.*, 2025], in which they introduced "a multimodal radio frequency dataset" for both human behavior recognition and emotion analysis. By conducting a series of experiments on 44 participants, they collected data from 13 radars, 8 RFID tags, LoRa devices and infrared cameras in 4 different campaigns that each focused on a different aspect of human behavior. We will be interested in this paper on the third campaign that was built around 6 sentiment states: *depression*, *distraction*, *excitement*, *focus*, *relaxation* and *stress*.

3 Methodology

4 Experiment 1: Classification using Kinematic Features

5 Experiment 2

6 Experiment 3

7 Results

8 Discussion

Acknowledgments

References

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